Physics 610
Fall 2023
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Homework 3

Due Monday, September 25

1. Problem 4 to Chap. 6 of the textbook. "Resonant frequencies" = frequencies of the normal modes. Please start by writing the Cartesian coordinates of the particles as functions of the generalized coordinates $q = (\theta_1, \theta_2)$ and use these expressions to obtain the Lagrangian $L(q, \dot{q})$.

2. For a linear oscillator with a time-dependent frequency,

$$\ddot{q} + \omega^2(t) q = 0 \,,$$

one can define a linear operator (matrix) $A(t, 0)$ that maps initial data $X(0)$, where $X \equiv (q, \dot{q})$, to the corresponding solution at time t, as follows:

$$X(t) = A(t) X(0) \,.$$

When $\omega^2(t)$ is periodic, with a period T, an important role is played by the map during one period, $A(T)$.

Consider the special case when $\omega^2(t)$ is periodic and piecewise constant, of the form

$$\omega^2(t) = \begin{cases} \omega_0^2, & 0 < t < \tau \\ 0, & \tau < t < T \end{cases}$$

(and repeats periodically with period T for $t > T$), with $\omega_0^2 > 0$. Compute the matrix $A(T)$ for this case. As a check of your result, verify that the map is volume-preserving: $\det A(T) = 1$.